

**A Report on the Implementation and Analysis of Morse Code Generation Using an
8051 Microcontroller with Dual DAC Interface and Real-Time Waveform Observation
on a Digital Storage Oscilloscope**

BY

VEMULA LAKSHMI ANUSHKA

ABSTRACT

This project presents the hardware implementation and analysis of **Morse code generation** using an **8051 microcontroller** with a **Dual DAC interface**, and the real-time observation of the resulting waveforms on a **Digital Storage Oscilloscope (DSO)**. The system is programmed to convert the name “**PRIYA**” into Morse code by generating standardized **DOT** and **DASH** pulses. These pulses are produced as digital patterns on **Port-1**, which are then converted into corresponding **analog voltage levels** through the **Dual DAC module**. The program ensures accurate timing by incorporating **unit delays** for DOTs (1 unit), DASHes (3 units), **intra-element gaps**, **letter gaps**, and **word gaps**, as defined in Morse communication standards. The DAC outputs a **full-scale voltage** ($0xFF \rightarrow V_{max}$) for high pulses and **zero voltage** ($0x00 \rightarrow 0V$) for gaps, enabling clear analog pulse generation.

The output waveform is captured on the **DSO**, where the DOT, DASH, and inter-symbol spacing are clearly visible, validating both the **timing logic** in the code and the correct operation of the **DAC-based signal conditioning**. The experimental results successfully demonstrate the translation of **digital Morse sequences** into observable **analog pulses**, confirming the reliability of the designed embedded system. This project highlights precise **timing control**, **digital-to-analog conversion**, and practical **waveform analysis** using low-cost educational hardware.

AIM:

To design and implement a Morse code generation system using an 8051 microcontroller and Dual DAC interface, and to observe and analyze the corresponding analog DOT–DASH waveforms on a Digital Storage Oscilloscope (DSO) for the given input sequence “PRIYA.”

REQUIREMENTS:

- 8051 Microcontroller Development Board
- Dual DAC Module (8-bit digital-to-analog converter)
- Digital Storage Oscilloscope (DSO)
- Keil µVision IDE (for assembly and HEX file generation)

THEORY:

1. Morse Code

Morse code represents characters using DOT (·) and DASH (–) sequences.

- DOT = 1 time unit
- DASH = 3 time units
- Intra-element gap = 1 unit
- Letter gap = 3 units
- Word gap = 7 units

2. 8051 Microcontroller

- The **8051** is an 8-bit microcontroller with four parallel ports.
- In this project, **Port-1** is used to output Morse pulses as digital HIGH (0xFF) and LOW (0x00) signals based on software delays.

3. Dual DAC Interface

The **Dual DAC** converts 8-bit digital values from Port-1 into corresponding **analog voltages**.

- 0xFF → maximum analog voltage
- 0x00 → 0V

This enables smooth analog waveform generation.

4. Software Delay Generation

DOT, DASH, and spacing are created using **UNIT_DELAY** loops.

These delays ensure correct Morse timing for the characters in “PRIYA.”

5. DSO Waveform Analysis

A **Digital Storage Oscilloscope** displays the analog output:

- Short pulse → DOT
- Long pulse → DASH
- Gaps → spacing between letters/words

This verifies the correct Morse encoding and timing.

MORSE CODES:

Morse Code

A	• —	P	• — — •	1	• — — — —
B	— • • •	Q	— — • •	2	• • — — —
C	— • — •	R	• — • •	3	• • • — —
D	— • • •	S	• • • •	4	• • • • —
E	•	T	—	5	• • • • •
F	• • — • •	U	• • — —	6	— • • • •
G	— — — •	V	• • • — —	7	— — — • •
H	• • • • •	W	• — — —	8	— — — • • •
I	• •	X	— • • • —	9	— — — — •
J	— •	Y	— • — — —	0	— — — — —
K	— • — —	Z	— — • • •		
L	• — — • •				
M	— — —	.	• — • — — —	?	• • — — — • •
N	— • •	,	— — • • — —	/	— • • — — •
O	— — — —	!	— • — — • — —	=	— • • • — —

PROCEDURE:

1. Power the 8051 development board using a regulated 5V supply and ensure the reset and crystal oscillator circuits are functioning properly.
2. Write the Morse code generation program in Keil μ Vision, including DOT, DASH, intra-element gap, letter gap, and word gap routines.
3. Assemble and compile the program to generate the HEX file, ensuring no errors are present in the code.
4. Burn the HEX file into the 8051 microcontroller using a USB programmer (USBASP/PROGISP/FlashMagic).
5. Connect PORT1 (P1.0–P1.7) of the 8051 to the corresponding digital input pins (D0–D7) of the Dual DAC module.
6. Connect the DAC output pin (analog OUT) to Channel 1 of the Digital Storage Oscilloscope (DSO) and connect common ground between 8051, DAC, and DSO.
7. Switch ON the setup, allowing the 8051 to start executing the Morse code program and generate the DOT–DASH pulse pattern digitally.
8. Observe the DAC analog output on the DSO in time-domain view, adjusting volts/div and time/div settings until the pulses become clearly visible.
9. Identify the waveform sections corresponding to DOT (short pulses), DASH (long pulses), intra-letter spacing, and letter/word gaps for the name "PRIYA".
10. Record and analyze the waveform to verify that the generated analog signal accurately matches the Morse code timing pattern defined in the program.

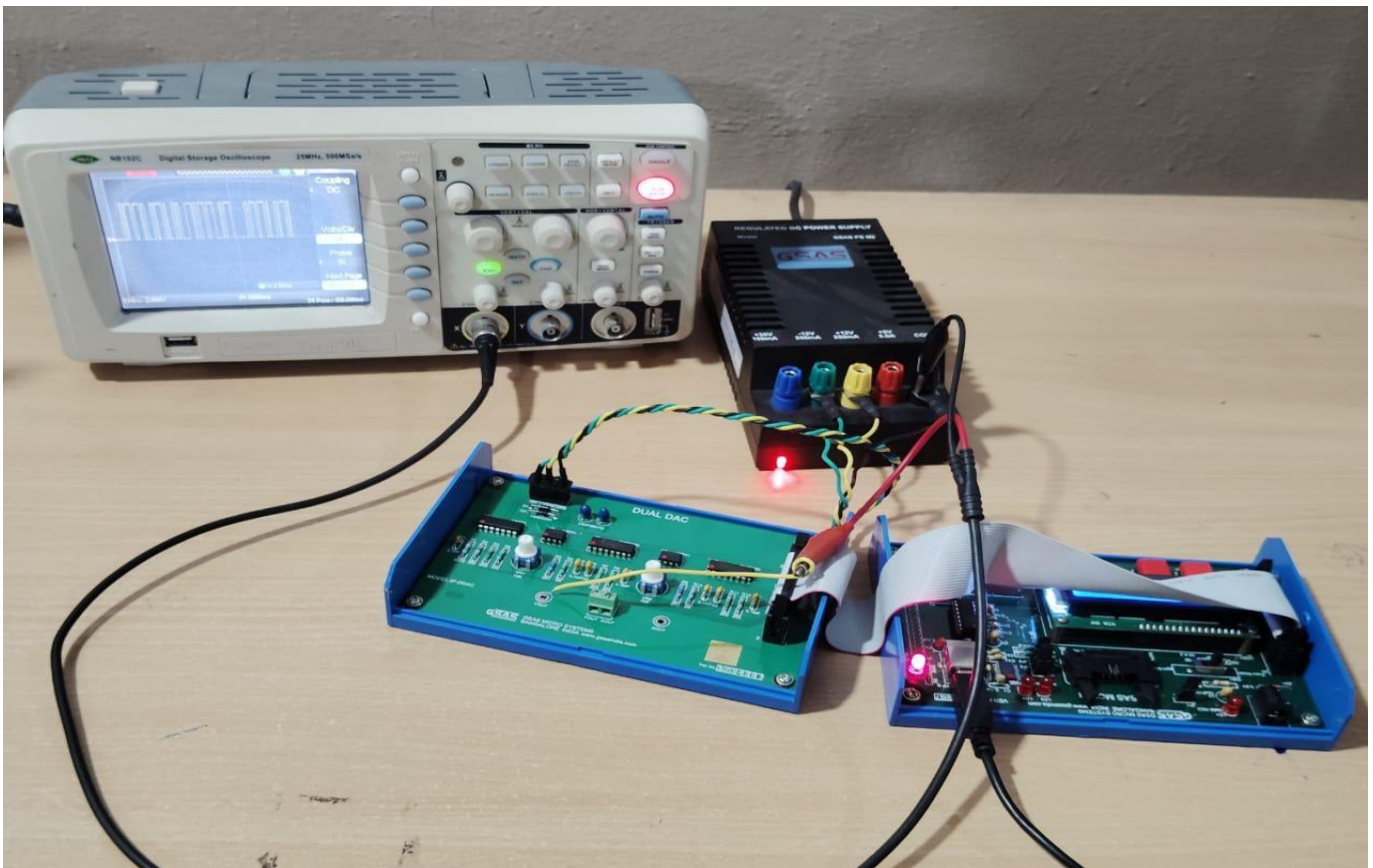
OBSERVATIONS:

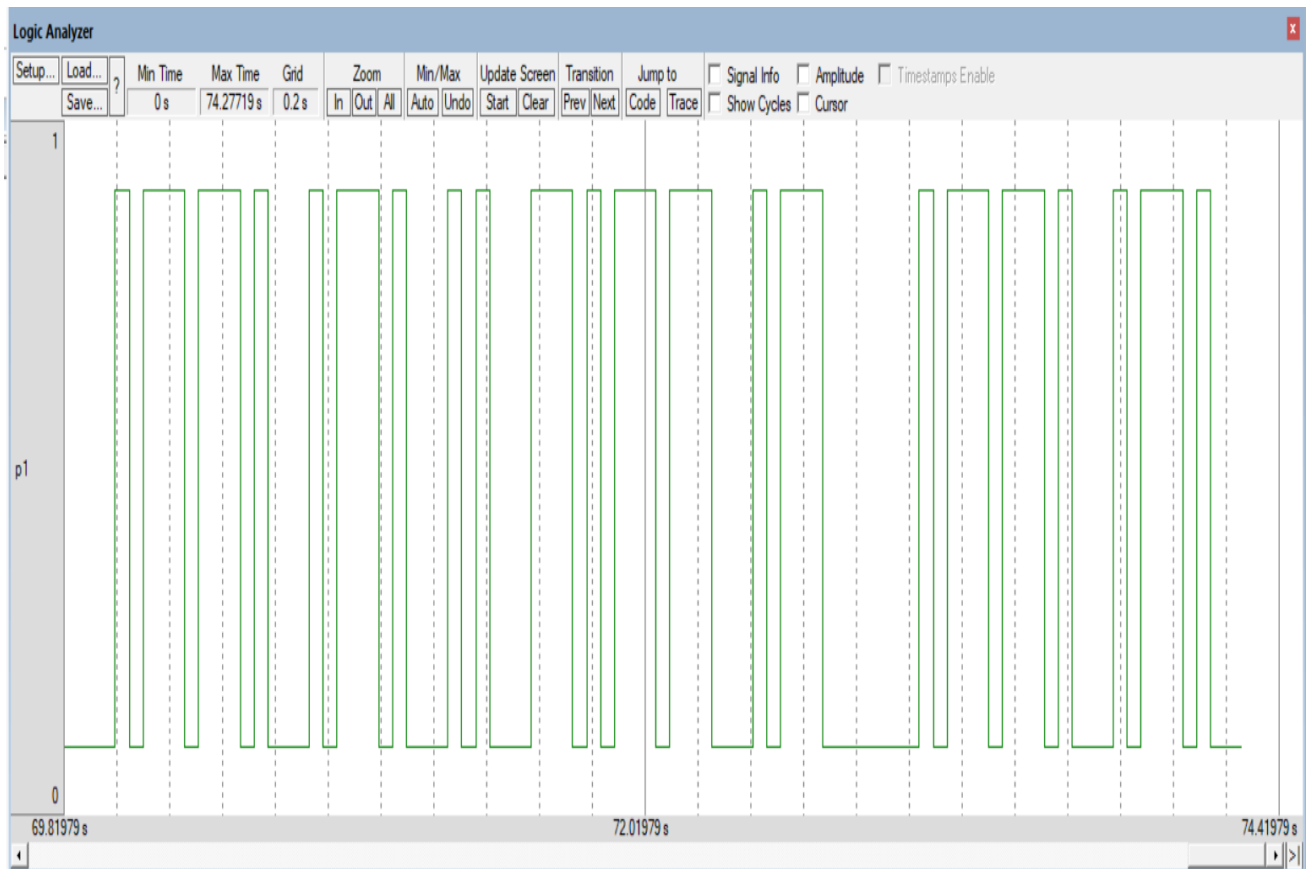
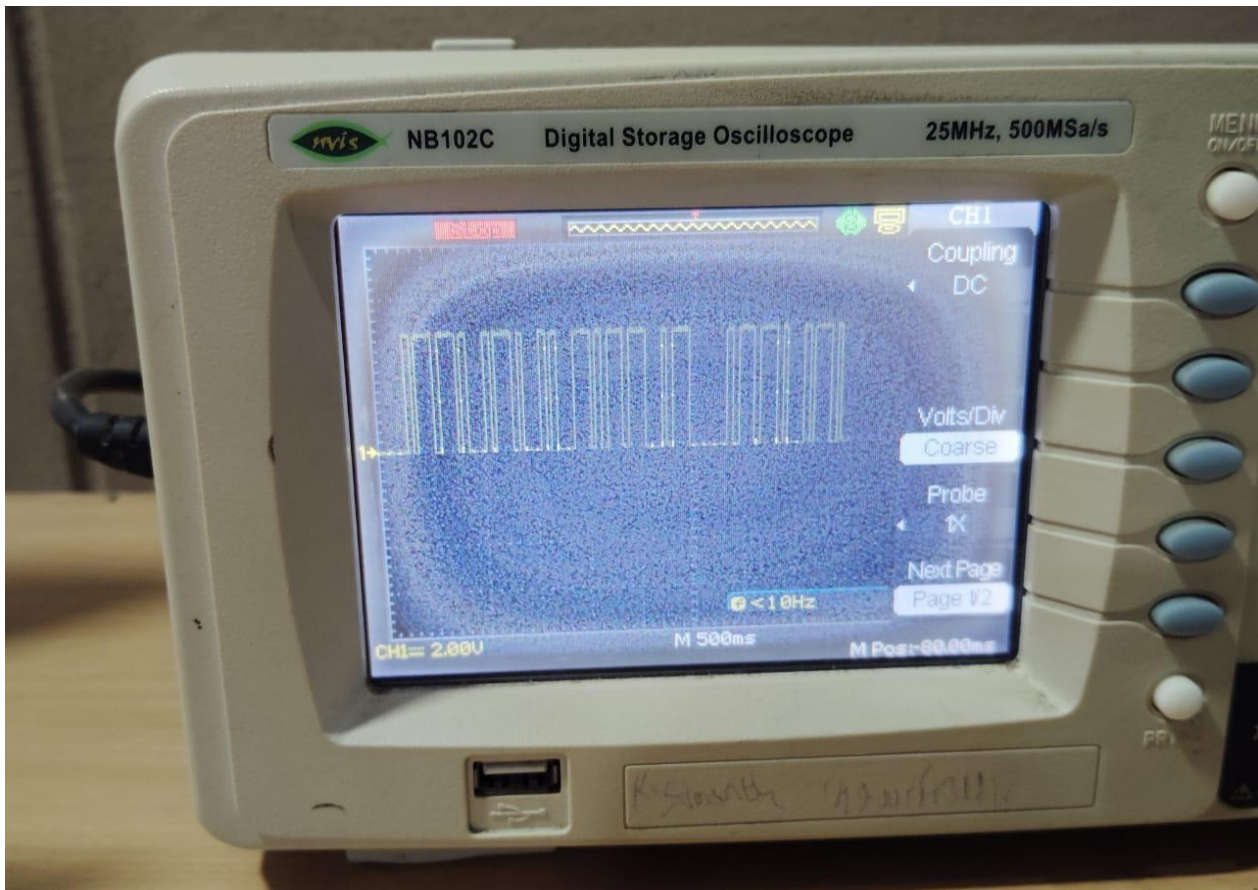
Morse Element	Code Representation	Expected DAC Output	Observed DSO Waveform
DOT	· (1 unit ON)	Short high pulse (V _{max})	Short-duration analog pulse observed
DASH	– (3 units ON)	Long high pulse (V _{max})	Long-duration analog pulse observed
Intra-element Gap	1 unit OFF	0V for 1 unit	Narrow low-level spacing between pulses
Letter Gap	3 units OFF	0V for longer duration	Clear separation between letters
Word Gap	7 units OFF	Extended 0V duration	Large gap before name repeats

Character-wise Observation (Name: PRIYA)

Letter	Morse Code	Expected Pattern	Observed Pattern on DSO
P	· - - ·	Short → Long → Long → Short	Visible pattern with correct spacing
R	· - ·	Short → Long → Short	All pulses detected clearly
I	· ·	Short → Short	Two short equal pulses observed
Y	- · - -	Long → Short → Long → Long	Distinct long & short pulses seen
A	· -	Short → Long	Accurate DOT-DASH sequence

RESULT:





The Morse code for the name “**PRIYA**” was successfully generated using the **8051 microcontroller** and converted into an analog waveform through the **Dual DAC interface**. The DOT and DASH pulses, along with the letter and word gaps, were clearly observed on the **Digital Storage Oscilloscope (DSO)**, matching the expected Morse code timing pattern.

CONCLUSION:

The Morse code generation system using the **8051 microcontroller** and **Dual DAC interface** was successfully designed and implemented. The programmed DOT–DASH sequences for the name “**PRIYA**” were accurately converted into analog pulses and clearly visualized on the **Digital Storage Oscilloscope (DSO)**. The observed waveforms matched the expected Morse timing, confirming proper DAC operation, correct delay generation, and reliable functioning of the embedded system. This project demonstrates the effective use of microcontroller-based digital signaling and DAC-based analog waveform generation.